



Fig. 6: Four-fibre bi-directional ring protection

an entire virtual ring is required for each path. In a BLSR, every link can carry both the working and protection traffic at the same time.

BLSR has two variants, namely, four-fibre BLSR and two-fibre BLSR. In a two-fibre BLSR, traffic is sent over both the fibres by utilising only half the capacity on each fibre and keeping rest half of the capacity reserved for protection.

In four-fibre BLSR, two fibres are

used as working line and the other two are used for protection (Fig. 6). Each pair of fibres transports working and protection channels.

This results in 1:1 protection, that is, 100 per cent redundancy. This improved protection is coupled with relatively high costs.

Ring and span switching mechanisms are used in the event of failure. In span switching, when the source or destination on a

link fails, traffic gets routed onto the protection fibre between the two nodes on the same link. And when a fibre or cable-cut occurs, service is restored using the ring switching mechanism. Two-fibre BLSRs also benefit from the ring switching but cannot perform span switching like a four-fibre BLSR.

Let us take an example of network failure in four-fibre bidirectional ring. If a fault occurs between neighbouring elements B and C, network element

C triggers protection switching and controls network element B by means of K1 and K2 bytes of the SDH system and traffic is re-routed as shown in Fig. 6.

## Conclusion

Network failures are very crucial and have always been a concern of utmost importance. Such failures may result in heavy losses of traffic, leading to complete service outage. In order to safeguard networks from failures, automatic protection switching mechanisms are being widely deployed in long-haul networks as well as in inter-office networks due to their high-efficiency capabilities coupled with very-low (less than 50ms) restoration time. ●

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